



NIRVANA: Structured Pruning Reimagined for Large Language Model Compression

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01 BACKGROUND

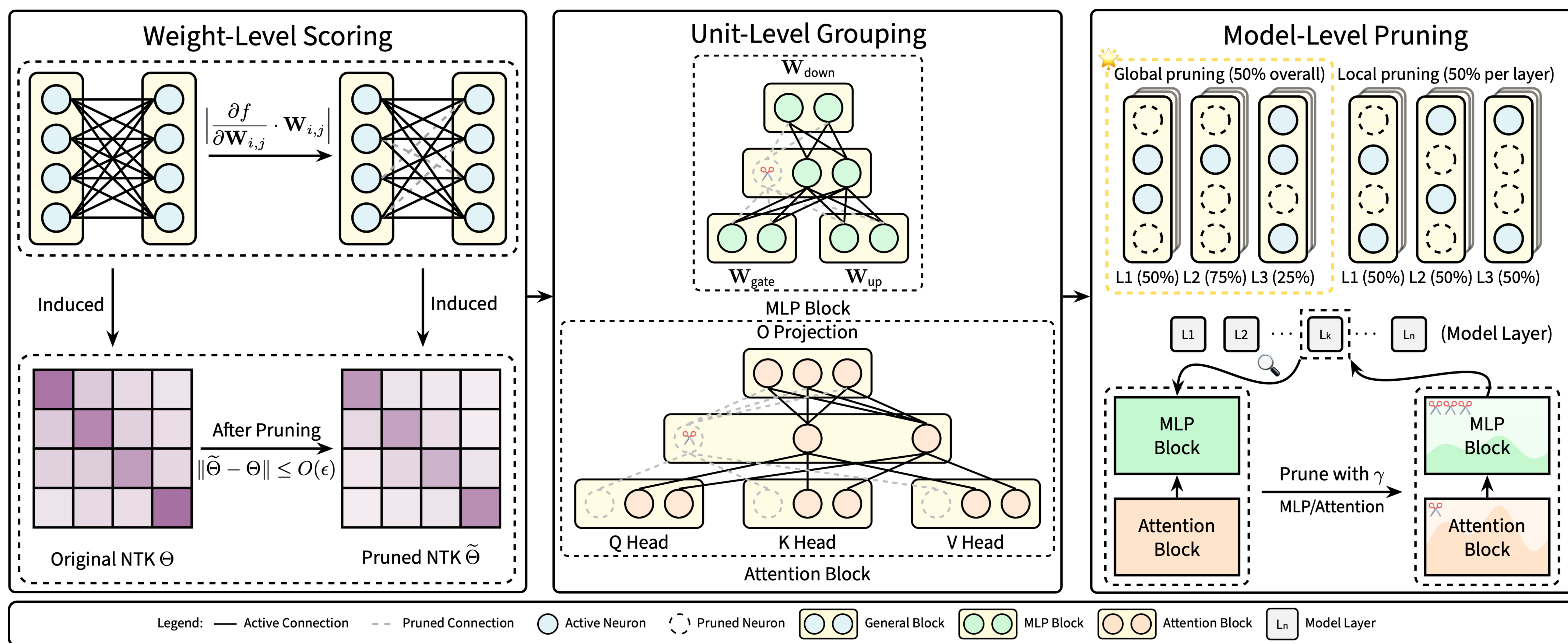
LLMs are expensive to deploy. **Structured pruning** removes entire attention heads and MLP neurons, shrinking model dimensions for acceleration on standard hardware.

02 MOTIVATION

How do we prune without losing capability? Reasoning and code generation can degrade sharply; layers, attention, and MLP units need different pruning budgets.

03 OUTPUT-BASED STRUCTURED PRUNING

Score weights by their output impact, group dependent neurons or heads, then prune whole structures.



Flexible attention / MLP allocation

Across layers: global ranking, nonuniform sparsity.

Across modules: use $v_{MLP} = \gamma v_{Attn}$ to balance pruning between attention heads and MLP neurons.

Output-based calibration

Select the calibration batch with the smallest KL divergence between dense and pruned outputs on a fixed anchor set.

04 RESULTS FROM THE PAPER

Llama3.1-8B: before recovery and after LoRA, across sparsity levels.

Sparsity	Method	WikiT ↓	PTB ↓	LambD ↓	ARC-e	WinoG	HellaS	SVAMP	MBPP*	Avg Acc	#Param
0%	Llama3.1-8B	8.50	14.02	20.09	81.52	73.56	78.90	72.67	48.60	71.05	8.03B
20%	LLM-Pruner	17.45	27.89	28.33	69.57	66.06	65.89	22.33	4.40	45.65	6.73B
	SliceGPT	21.10	118.79	252.46	53.70	61.61	51.40	0.00	0.00	33.34	6.77B
	FLAP	20.88	31.35	31.72	60.35	66.22	59.03	16.33	3.40	41.07	6.61B
	Tyr-the-Pruner	19.77	38.11	31.11	68.03	66.46	67.27	19.67	4.00	45.09	6.66B
	NIRVANA (ours)	13.38	19.77	26.20	<u>68.52</u>	67.40	<u>66.75</u>	49.00	23.80	55.09	6.62B
w/ LoRA [†]	LLM-Pruner	12.57	19.77	25.00	74.92	68.27	74.06	33.33	25.20	55.16	6.75B
	SliceGPT	49.81	87.25	81.64	61.53	61.56	59.91	0.00	0.60	36.72	6.58B
	FLAP	16.14	25.00	30.63	70.71	63.54	67.58	34.33	15.00	50.23	6.63B
	Tyr-the-Pruner	16.65	30.15	29.68	75.67	68.51	72.40	39.67	21.00	55.45	6.68B
	NIRVANA (ours)	12.37	18.58	25.00	76.98	69.54	74.36	56.67	33.00	62.11	6.64B
25%	LLM-Pruner	22.41	39.33	33.64	<u>60.06</u>	56.35	48.16	<u>16.33</u>	0.00	36.18	6.29B
	SliceGPT	22.19	155.54	280.54	44.83	54.15	39.91	0.00	0.00	27.78	6.57B
	FLAP	24.71	37.38	34.37	56.06	<u>65.27</u>	55.60	11.00	2.60	38.11	6.33B
	Tyr-the-Pruner	22.76	<u>31.11</u>	<u>31.60</u>	65.03	63.85	<u>61.82</u>	13.33	<u>3.20</u>	<u>41.45</u>	6.35B
	NIRVANA (ours)	15.64	22.41	28.77	<u>64.65</u>	65.56	62.21	27.33	16.00	47.15	6.28B
33%	LLM-Pruner	30.15	<u>52.92</u>	<u>39.94</u>	51.85	55.80	42.78	<u>13.67</u>	0.00	32.82	5.83B
	SliceGPT	35.05	315.42	446.55	37.71	51.62	32.67	0.00	0.00	24.40	6.06B
	FLAP	35.39	54.60	43.11	46.00	60.54	48.48	10.33	0.00	33.07	5.73B
	Tyr-the-Pruner	30.63	54.60	43.19	<u>55.51</u>	56.75	<u>52.74</u>	7.67	0.00	<u>34.53</u>	5.87B
	NIRVANA (ours)	20.40	29.22	34.70	56.44	60.77	53.93	20.67	7.20	39.80	5.70B
50%	LLM-Pruner	215.94	356.02	196.62	31.19	49.01	28.82	7.33	0.00	23.27	4.55B
	SliceGPT	93.24	612.76	870.90	32.37	49.57	29.70	0.00	0.00	22.33	4.57B
	FLAP	<u>68.21</u>	<u>97.71</u>	61.87	36.45	<u>53.56</u>	<u>37.20</u>	<u>8.33</u>	0.00	<u>27.11</u>	4.50B
	Tyr-the-Pruner	123.04	196.62	115.58	38.64	50.59	35.39	5.33	0.00	25.99	4.72B
	NIRVANA (ours)	48.94	70.11	<u>67.95</u>	<u>37.54</u>	54.38	37.72	9.33	0.20	27.83	4.51B
w/ LoRA [†]	LLM-Pruner	45.26	92.87	70.12	48.32	52.41	43.51	4.67	0.00	29.78	4.55B
	SliceGPT	198.93	246.61	248.55	40.19	52.49	36.57	0.00	0.00	25.85	4.59B
	FLAP	<u>28.33</u>	<u>42.52</u>	<u>43.87</u>	50.76	58.09	<u>49.90</u>	<u>15.67</u>	0.00	<u>34.88</u>	4.52B
	Tyr-the-Pruner	54.60	81.96	81.96	<u>52.40</u>	54.06	<u>46.09</u>	13.33	0.00	33.18	4.74B
	NIRVANA (ours)	25.79	36.94	42.52	57.49	56.27	49.91	23.00	3.40	38.01	4.53B

* MBPP: 3-shot pass@1. † LoRA at 20% / 50% sparsity. Bold: best; underline: second-best.

23.8% vs. 4.4% MBPP NIRVANA vs. LLM-Pruner at 20% sparsity.

INFERENCE EFFICIENCY 50% target sparsity, single A100, batch 32

1.57× throughput 12.41 → 19.50 ktokens/s

MEMORY FOOTPRINT

17.95 → 11.27 GB Dense → NIRVANA

CONTACT

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Paper



Code



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